Instituto Superior de Engenharia do Porto

ALGAN	Test	
Course. LEI	**Date: 2024 – 11 – 16**	**75 minutes**

Name: _____ **Number**: _____ **Class**: _____

Part I (15 values)

1. (**1,5+1,5 val.**) Consider the matrices,

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{bmatrix} \quad , \quad \mathbf{B} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

   a) Calculate $\det(\mathbf{AB} - \mathbf{BA})$.

   b) Use Gauss-Jordan algorithm to calculate $\mathbf{B^{-1}}$.

2. (**3 val.**) Consider the matrix $\mathbf{A} = \begin{bmatrix} -2 & -a & 2 & a \\ 3 & -3 & a & 2a \\ a & -5 & 5 & 3a \\ 0 & 0 & 0 & 4a \end{bmatrix}$, where a is a real parameter.

   For which values of the parameter a is the matrix $\mathbf{A}$ invertible?

3. (**3+1 val.**) Consider the system of linear equations $\begin{cases} x + 2y + z = a \\ x + (a^2+1)y + az = 2a \\ -x - 2a^2y - az = -3a + b - 1 \end{cases}$ where a and b are real parameters.

   a) Classify the SLE in order of the parameters a and b.

   b) Solve the system for $a = -1$ and $b = 0$.

4. (**1+3+1 val.**) Consider the following system of linear equations (SLE).

$$\begin{cases} x_1 + 2x_2 + 3x_3 + x_4 = 1 \\ x_1 + x_2 + x_3 + x_5 = 2 \\ x_1 - x_2 - x_3 + x_6 = 3 \end{cases}$$

   a) Show that $(\frac{5-a}{2}, a, -\frac{a+1}{2}, 0, 0, a)$ is a solution of the SLE for all $a \in \mathbb{R}$.

   b) Find the solution set of the SLE.

   c) Find, if possible, a solution $(s_1, s_2, s_3, s_4, s_5, s_6)$ of the SLE that simultaneously satisfies the following conditions: $s_4 = -1$, $s_6 = -1$ and $\sum_{i=1}^{6} s_i = 0$.

Part II (5 values)

To each question corresponds a unique correct answer. The answers must be presented in the following grid with one of the options (A,B,C or D). If you do not answer or give more than one answer to a question, you will have **zero** values in that question. Each correct answer worth **1 value**, and each wrong answer will have a penalty of **1/3** values.

Question	5	6	7	8	9	NA	W	C	Total
Answer									

. . . . . .

5. Let $\mathbf{A} \in \mathcal{M}_5$ be a matrix such that $\mathbf{A^{-2}} = -\mathbf{A^T}$. Then we can conclude that:

   A. $\det(\mathbf{A}) = 1$; B. $\det(\mathbf{A}) = 0$; C. $\det(\mathbf{A}) = 2$; D. None of the others is correct.

6. Consider the matrices

$$\mathbf{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 6a & 6c & 12b \\ d & f & 2e \\ g & i & 2h \end{bmatrix}$$

   Then we can conclude that:

   A. $\det(\mathbf{A}) = \det(\mathbf{B})$; B. $\det(\mathbf{A}) = -12\det(\mathbf{B})$; C. $\det(\mathbf{A}) = 6\det(\mathbf{B})$; D. None of the others is correct.

7. Let $\mathbf{A} \in \mathcal{M}_{14}$ be a matrix such that: $\mathbf{A}^8 = \mathbf{I_{14}}$. Then we can conclude that:

   A. $\mathbf{A}$ não é invertível; B. $\mathbf{A}^{-6} = \mathbf{A}^2$; C. $\det(\mathbf{A}) = \frac{1}{8}$; D. None of the other options can be concluded.

8. Let $\mathbf{A} \in \mathcal{M}_4$ be such that $\mathbf{A^3} + \mathbf{A^2} = -\mathbf{I_4}$

   Then we can conclude that:

   A. $\det(\mathbf{A}) = \pm\frac{1}{\sqrt{\det(\mathbf{A}+\mathbf{I_4})}}$; B. $\det(\mathbf{A^2}) = -\det(\mathbf{A}+\mathbf{I_4})$; C. $\det(\mathbf{A}+\mathbf{I_4}) \leq 0$; D. None of the others is correct.

9. Consider the following matrix equation $((\mathbf{AX})^{-1}\mathbf{B})^T = \mathbf{C}$, where the matrices involved are square, with the same order, and invertible. Then we have:

   A. $\mathbf{X} = \mathbf{A}^{-1}\mathbf{B}(\mathbf{C}^{-1})^T$;

   B. $\mathbf{X} = \mathbf{B}^T\mathbf{A}^{-1}\mathbf{C}^T$;

   C. $\mathbf{X} = \mathbf{A}^{-1}(\mathbf{B}^{-1}\mathbf{C})^T$;

   D. None of the others is correct.